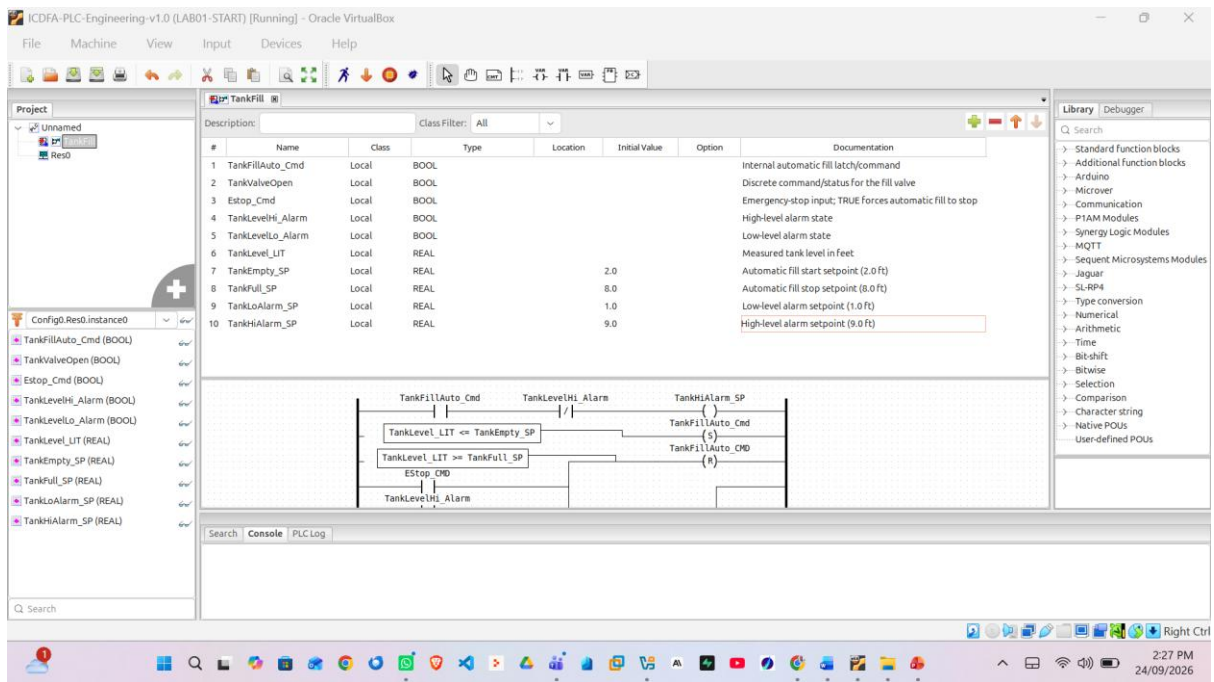
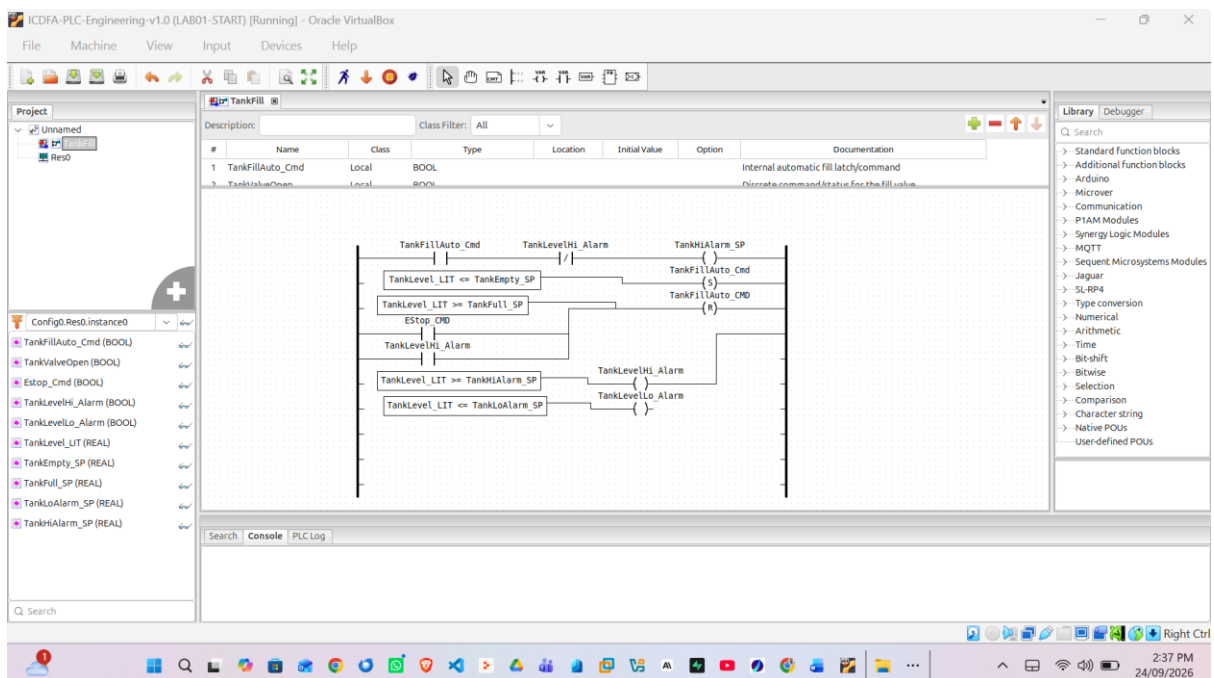




Variable/tag table.



Ladder program.

A six-scenario logic test table showing inputs/conditions and expected outputs.

Scenario	Inputs / Conditions	Expected Outputs
1. Empty Tank	TankLevel_LIT $\leq$ TankEmpty_SP ($\leq$ 2.0 ft), Estop_Cmd = FALSE	TankFillAuto_Cmd is SET, TankValveOpen becomes TRUE, filling process starts.
2. Normal Filling	Tank level is between 2.0 ft and 8.0 ft, TankFillAuto_Cmd = TRUE, TankLevelHi_Alarm = FALSE	TankValveOpen remains TRUE and the tank continues filling normally.
3. Full Tank	TankLevel_LIT $\geq$ TankFull_SP ($\geq$ 8.0 ft)	TankFillAuto_Cmd is RESET and TankValveOpen becomes FALSE, stopping the filling process.
4. High Alarm Condition	TankLevel_LIT $\geq$ TankHiAlarm_SP ($\geq$ 9.0 ft)	TankLevelHi_Alarm becomes TRUE and automatic filling is stopped for safety.
5. Low Alarm Condition	TankLevel_LIT $\leq$ TankLoAlarm_SP ($\leq$ 1.0 ft)	TankLevelLo_Alarm becomes TRUE to indicate a low tank level condition.
6. Emergency Stop	Estop_Cmd = TRUE regardless of tank level	TankFillAuto_Cmd is RESET, TankValveOpen becomes FALSE, and the automatic filling operation stops immediately.

Safety Explanation: Emergency Stop Override

An emergency-stop condition must override normal automatic control to ensure the safety of personnel, equipment, and the industrial process. When an emergency stop signal is activated, the PLC must immediately stop the automatic operation regardless of the current tank level or filling condition. This prevents unsafe situations such as overfilling, equipment damage, or hazardous operating conditions.

In the tank-fill control system, the **Estop_Cmd** input has priority over the automatic filling logic. When the emergency stop is TRUE, the automatic fill command is reset and the fill valve is closed, ensuring that the process moves to a safe state before normal operation can resume.

Knowledge Check

1. Why are BOOL and REAL appropriate data types for this project?

BOOL and REAL data types are appropriate because the project contains both discrete control signals and continuous measurements. BOOL variables are used for ON/OFF conditions such as the fill command, valve status, alarms, and emergency stop, while REAL variables are used for tank level measurements and setpoint values that require numerical values.

2. What is the purpose of a SET/RESET pair for TankFillAuto_Cmd?

The SET/RESET pair is used to maintain and control the automatic filling command. The SET instruction activates TankFillAuto_Cmd when the tank level reaches the start condition, while the RESET instruction deactivates it when the tank reaches the full condition or when a safety condition such as an emergency stop occurs.

3. What should happen if the tank reaches 9 ft while the fill valve remains commanded open?

When the tank reaches 9 ft, the high-level alarm condition is triggered. The PLC should activate TankLevelHi_Alarm and stop the automatic filling operation by resetting the fill command and closing the fill valve to prevent overfilling.

4. Why should PLC logic be validated in simulation before use with physical equipment?

PLC logic should be validated in simulation to confirm that the control sequence, safety conditions, alarms, and outputs operate correctly before connecting to physical equipment. This reduces the risk of equipment damage, unsafe operation, and unexpected behaviour during real-world deployment.